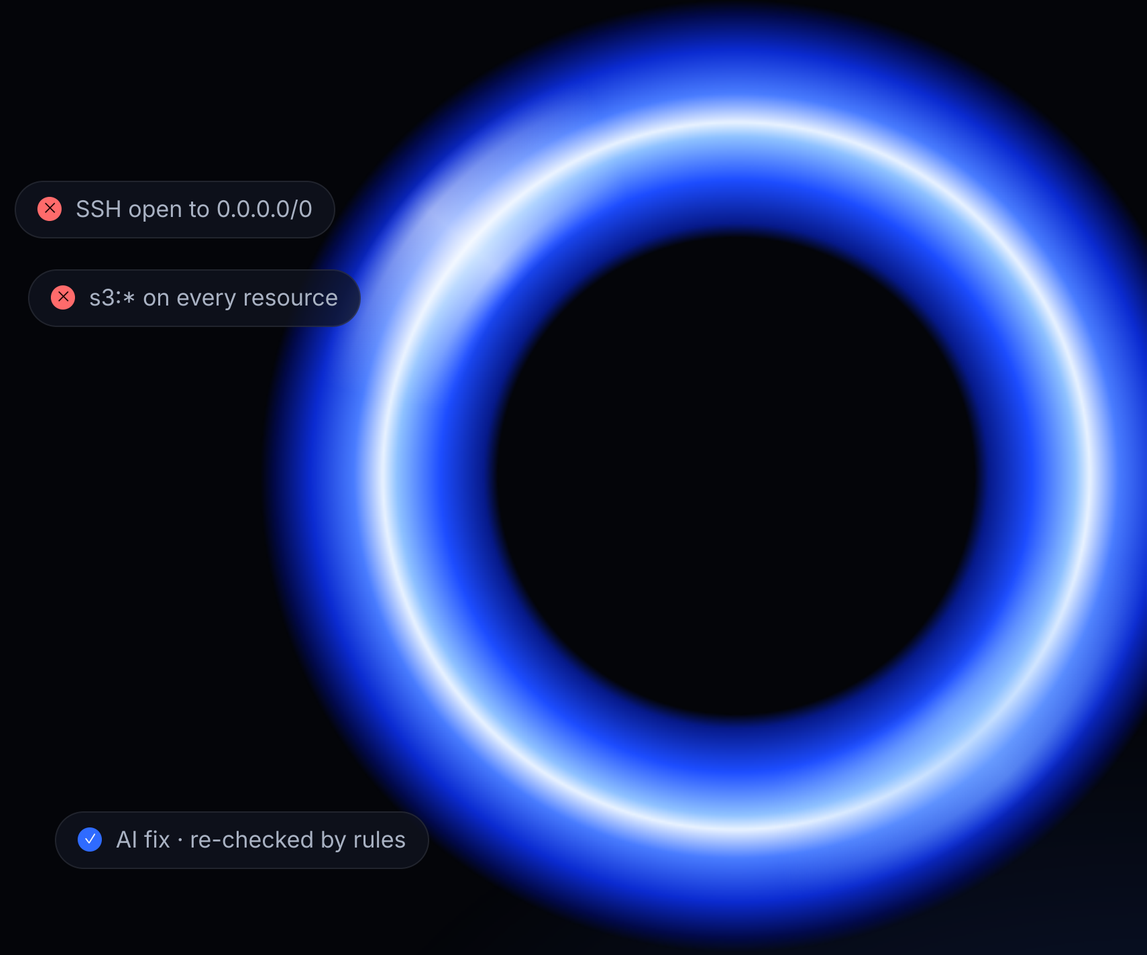
01 · PROJECT TITLE · TENTATIVE

Explainable Cloud Security Analyzer

with Attack-Path Analysis and AI-Assisted Remediation

A Terraform-based, pre-deployment analysis for AWS.



PRESENTED BY

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23001990 · Bachelor of Computer Science (Information Systems)

SUPERVISOR

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FYP Monitoring Presentation

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Tentative title. Each part maps to one capability of the analyzer.

What the title promises

Explainable Cloud Security Analyzer with Attack-Path Analysis and AI-Assisted Remediation

EXPLAINABLE 01

Every finding shows its evidence

The exact Terraform line, why it is risky, and how to fix it.

ATTACK-PATH ANALYSIS 02

Weaknesses are connected

Resources, identities and permissions linked into one potential path.

AI-ASSISTED REMEDIATION 03

AI suggests, you approve

A proposed Terraform fix, reviewed as a diff, then re-checked by the rules.

AWS · Terraform · S3 · Security Groups · IAM · Before deployment · Never runs terraform apply

02

01 Cloud infrastructure is now written as code with Terraform.

02 One wrong line is deployed every time the file is applied.

03 Scanners list findings one by one, not how they connect.

main.tf · excerpt

```
3        from_port   = 22
6        cidr_blocks = ["0.0.0.0/0"]                    SSH open to the internet
13     vpc_security_group_ids = [aws_security_group.web.id]
14     iam_instance_profile   = aws_iam_instance_profile.app.name
29       Action = "s3:*", Resource = "*"                every S3 action
33   resource "aws_s3_bucket" "records" {               no public access block
```

17

failed Checkov checks on one 35-line file, listed one by one

3 → 1

three findings that form one potential attack path

0

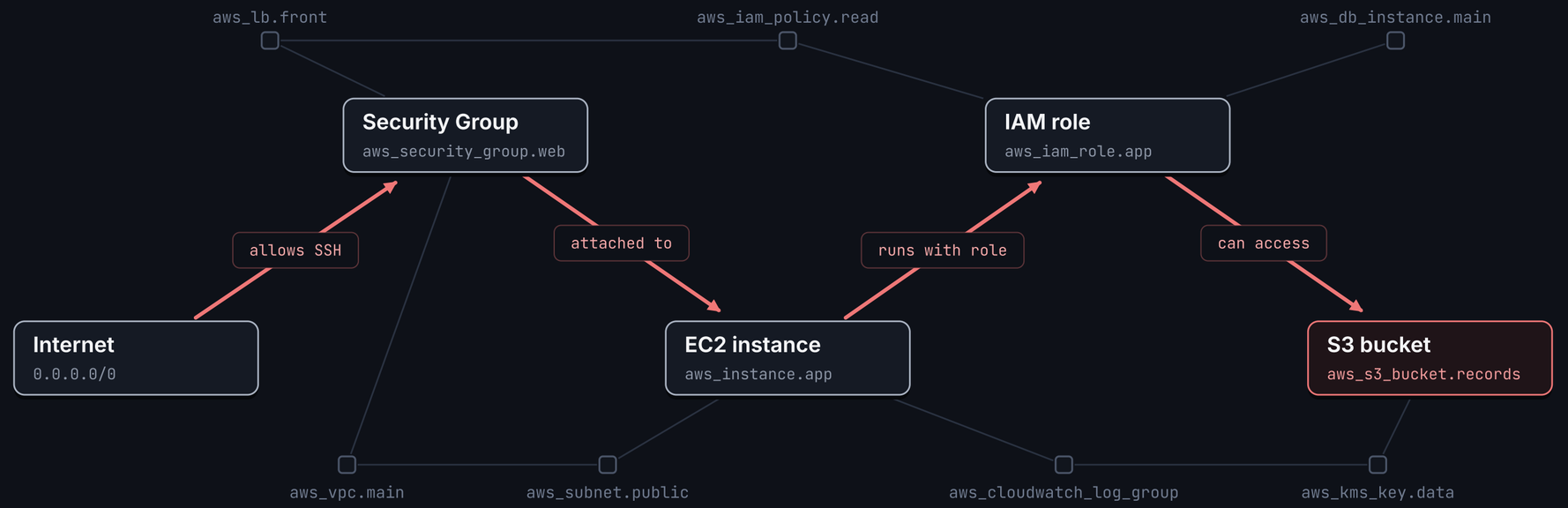
automatic deployments: analysis runs before terraform apply

Rules detect. AI suggests. You approve. Rules re-check.

How the analyzer works

01	02	03	04	05	06	07
Terraform	**Plan → JSON**	**Detect**	**AI-assisted fix**	**Review**	**Re-scan**	**Deploy**
write main.tf	terraform show -json	rules · severity · paths	proposed patch	diff · accept or reject	fmt · validate · rules	your decision
YOU	ANALYZER	ANALYZER	AI MODEL	YOU	ANALYZER	YOU

ONE POTENTIAL ATTACK PATH, FROM MAIN.TF



potential attack path — other relationships in main.tf

Ranked first: one combined risk, above its separate findings. A potential path from configuration evidence, not a confirmed exploit.

RULE CATALOGUE

10

declarative rules, each mapped to CIS / AWS guidance.

- S3 × 4
- Security Groups × 3
- IAM × 3

Each objective answers one or more problem statements.

Four objectives, one analyzer

01



Detect

Identify selected AWS misconfigurations in Terraform with 10 rule-based checks.

Answers PS1

02

Connect

Model resources, identities and permissions to find potential attack paths.

Answers PS1

03

Explain, prioritise & fix

Evidence-linked findings, contextual ranking, and AI-suggested fixes that the rules re-check.

Answers PS2 · PS3

04

Evaluate

Labelled Terraform scenarios, compared with Checkov; optional read-only AWS check.

Measures O1–O3

8–10 labelled scenarios

≥3 clean · 2 tricky-but-safe · ≥2 attack paths

Precision · recall · F1

Fixes resolved on re-scan

What existing scanners report, and what they leave to the reader.

Three findings. One hidden path.

TODAY · ISOLATED FINDINGS

- SSH open to 0.0.0.0/0
- s3:* on every resource
- No public access block

NEEDED · CONNECTED VIEW



Detect

PS1

Limited contextual analysis of interconnected IaC configurations

Rules check one resource at a time, not how they combine.

Understand

PS2

Limited explainability of interconnected security findings

A finding names the weakness, not its evidence or impact.

Prioritise

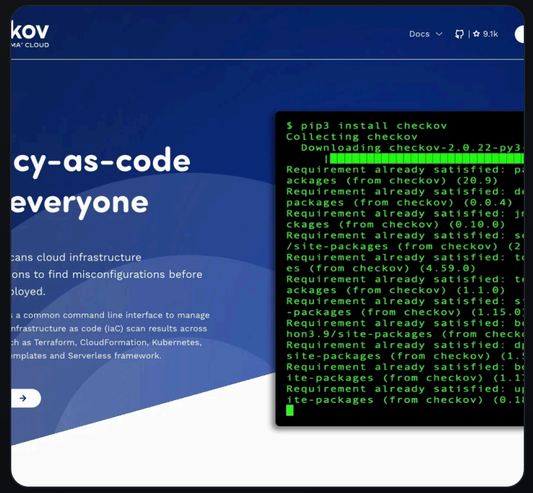
PS3

Limited contextual prioritisation of combined cloud security risks

Scored one by one, combined risk stays hidden.

Official pages, captured 11 Nov 2026. Click a card to open the source.

What already exists



Checkov

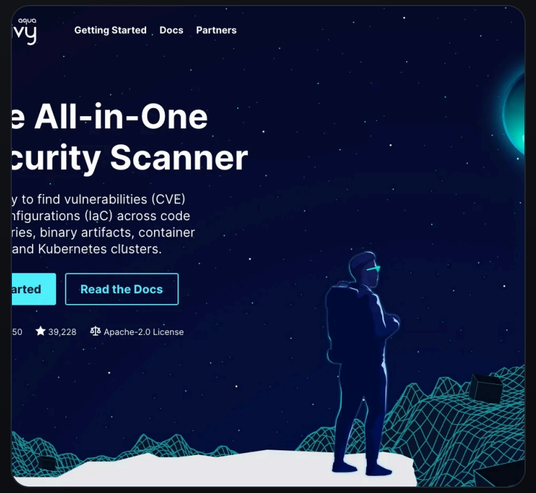
IaC · pre-deploy

STRENGTH

Hundreds of built-in Terraform policies.

GAP

Findings reported one by one.



Trivy

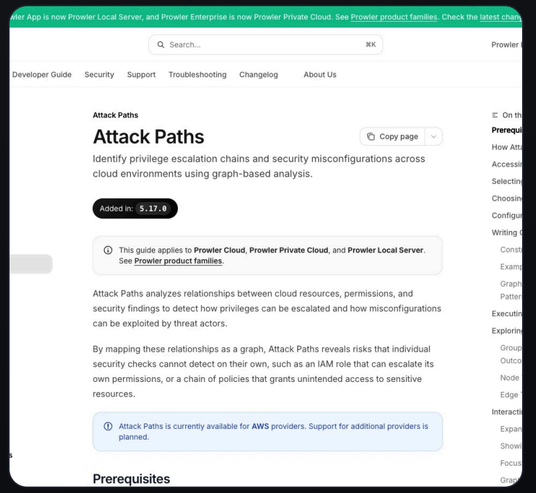
IaC · images

STRENGTH

Vulnerabilities, misconfigurations and secrets.

GAP

No relationship analysis in sources reviewed.



Prowler

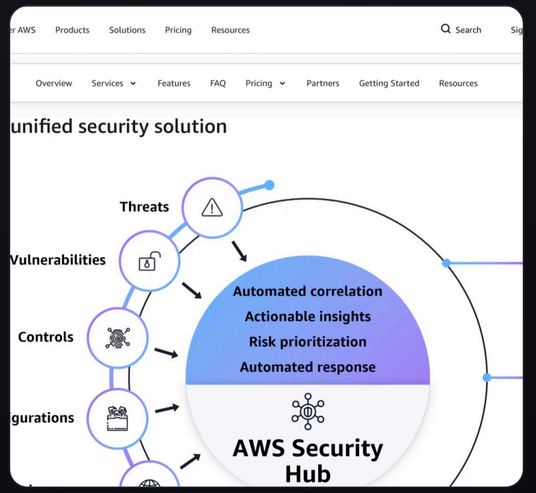
live account

STRENGTH

Graph-based attack paths for AWS.

GAP

Needs a deployed account.



Security Hub

live account

STRENGTH

Correlates and prioritises AWS risks.

GAP

Works after deployment.

Three verified papers (DOI checked). Click to open.

Research that supports it



INFRASTRUCTURE-AS-CODE SECURITY AND DRIFT DETECTION USING LLMs

Movva et al. · 2025

IaC Security and Drift Detection Using LLMs

Context-dependent misconfigurations are missed by fixed rules.

Limit its 31% gain is paper-specific



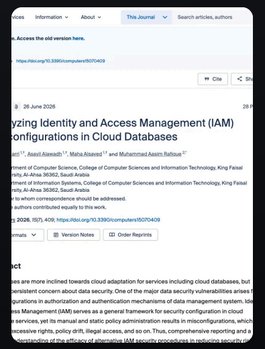
STATIC ANALYSIS AND LLM-BASED REVIEW FOR TERRAFORM SECURITY AND COST ISSUES: AN EXPLORATORY COMPARATIVE STUDY

Keerthana · 2026

Static Analysis and LLM-Based Review for Terraform

LLM review adds explanations and suggested fixes to Checkov.

Limit exploratory, 12 Terraform files



Almarri et al. · 2026

IAM Misconfigurations in Cloud Databases

Excess permissions lead to privilege escalation: IAM is in scope.

Limit focused on cloud databases

The gap this project fills

Before deployment
Terraform plan, not a live account

Connected paths
findings linked by evidence

Explained & ranked
why it matters, what first

AI fix, re-checked
suggested, approved, re-scanned

Status as of November 2026.

Who shapes the work

Supervisor

FA

Dr. Fairuz Amalina Narudin

FYP supervisor

ROLE

Academic guidance on scope, rule selection and evaluation design.

Collaborator

LC

Lejiend Community

Tech community · collaborator

INPUT SOUGHT

Practitioner feedback on test scenarios and on how clear the explanations are.

lejiend.com ↗ Owner ↗

Responded

FH

Farhan Helmy

AI & full-stack engineer

INPUT GIVEN

Suggested the AI-assisted remediation feature.

INPUT SOUGHT

Feedback on the AI fix workflow and the report.

farhanhelmy.com ↗ LinkedIn ↗

Potential

+

Cloud security practitioners

Industry perspective

INPUT SOUGHT

Common real-world AWS misconfiguration patterns.

Done · AI fix idea from industry input

Planned · practitioners review test scenarios

Planned · clarity check of explanations

REFERENCES

DOIs verified against Crossref, 11 Nov 2026.

Sources

RESEARCH PAPERS

Movva, D. K., Pullikallu, A. R., & Sugasi, M. K. (2025). Infrastructure-as-Code Security and Drift Detection Using LLMs. *International Journal of Computer Security, 2*(1), 9–25.

Keerthana, K. S. (2026). Static Analysis and LLM-Based Review for Terraform Security and Cost Issues: An Exploratory Comparative Study. *IJEAST, 11*(5), 95–99.

Almarri, A., Alawadh, A., Alsayed, M., & Rafique, M. A. (2026). Analyzing Identity and Access Management (IAM) Misconfigurations in Cloud Databases. *Computers, 15*(7), 409.

TOOLS, DOCUMENTATION & STANDARDS

Checkov · Trivy misconfiguration scanning

Prowler Attack Paths · AWS Security Hub attack path graph

Terraform JSON output format

CIS Amazon Web Services Foundations Benchmark v7.0.0

Thank you

Questions and feedback are welcome.

Detect · Connect · Explain · Prioritise · AI-assisted fix · Validate